THE WORLD COMES HERE

# TMS 2027

156th Annual Meeting & Exhibition

MARCH 14–18, 2027
ORLANDO WORLD CENTER MARRIOTT
ORLANDO, FLORIDA, USA

## JOIN US FOR THE FOLLOWING SYMPOSIUM

www.tms.org/TMS2027

### DATA-DRIVEN AND COMPUTATIONAL MATERIALS DESIGN

# AI-Enabled Materials Processing: Integrating Accelerated Experimental Workflows and Processing-Aware Machine Learning

Processing governs microstructure evolution, defect populations, and interfaces across metals, alloys, composites, and functional materials, thereby determining reliability, manufacturability, and performance. This symposium centers on the integration of artificial intelligence and machine learning directly with experimental materials processing, rather than on general AI-driven materials discovery. While many current AI and ML efforts emphasize composition or purely computational modeling, comparatively less attention has been given to treating processing history as a primary design variable. Advances in high-throughput manufacturing, accelerated characterization and testing, and real-time process monitoring now create opportunities to embed data-driven and physics-informed methods within experimental processing workflows.

The symposium focuses on process-aware microstructure and property control enabled by AI integrated with thermomechanical processing, heat treatment, casting, additive manufacturing, surface modification, and deposition-based routes. Contributions may be experimental, computational, or hybrid in nature, but should emphasize coupling AI with processing science to accelerate optimization and scale-up. Particular emphasis is placed on generating and structuring high-quality processing-microstructure-property datasets from manufacturing workflows and leveraging them to encode processing histories into predictive models of microstructure and properties. Topics include AI-assisted exploration of complex parameter spaces for processing; surrogate and reduced-order models for complex processing routes; uncertainty-aware and few-shot learning approaches for small or sparse datasets; Bayesian optimization for process tuning; and closed-loop experimental workflows that integrate synthesis, processing, characterization, testing, and iterative model-guided refinement.

The goal of this symposium is to bring together materials processors, metallurgists, manufacturing engineers, and data scientists working at the interface of experimental processing and data-driven optimization. By positioning processing as a central axis in AI-enabled materials development, this symposium aims to accelerate the translation of laboratory-scale insights into robust, manufacturable, and scalable materials systems. Topics of Interest Include, But Are Not Limited To: Integration of AI/ML with experimental processing for microstructure and property control, process optimization, and manufacturing scale-up High throughput and data-aware experimental design for systematic exploration of processing parameter spaces Processing-structure-property relationships supported by advanced and accelerated characterization, testing, and real-time process monitoring Bayesian optimization, adaptive experimentation, and closed-loop control in manufacturing-relevant environments Applications in structural alloys, biocompatible materials, additively manufactured components, coatings, batteries, and other processing-intensive systems Experimental advances in casting, rolling, extrusion, forging, heat treatment, additive manufacturing, and deposition-based processes

## DIVISION SPONSOR(S)

TMS Materials Processing and Manufacturing Division

## COMMITTEE SPONSOR(S)

TMS: Additive Manufacturing Committee; TMS: Computational Materials Science and Engineering Committee; TMS: Process Technology and Modeling Committee

## ORGANIZING COMMITTEE


- **Sreenivas Raguraman**, Johns Hopkins University
- **Maitreyee Sharma Priyadarshini**, Virginia Tech
- **Timothy Weihs**, Johns Hopkins University
- **Thomas Voisin**, Lawrence Livermore National Laboratory
- **Allison Beese**, Pennsylvania State University
- **Samantha Webster**, Colorado School of Mines


**QUESTIONS?** Contact **programming@tms.org**

**KEY DATES:**
**July 1, 2026**: Abstract Submissions Due
**October 2026**: Registration Opens
**January 31, 2027**: Discount Registration Deadline